Explicit irreducible representations of the Hecke algebras of the cubic arc-transitive catalogue

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Abstract

For every (G, H) coset pair and every graph in the catalogue of cubic arc-transitive graphs (`CATALOGUE.md`), we compute all irreducible representations of the Hecke algebra $\mathcal{H}(G, H) = \mathrm{End}_G \mathbb{C}[G/H]$ explicitly: spherical function tables for the Gelfand pairs, and exact matrix entries (over cyclotomic fields) for every irreducible representation of dimension ≥ 2 in the non-Gelfand cases. The computation independently re-derives the Meinhold-72 relations $A^2 = I$, $B^2 = 13I$, $AB + BA = -4I$, and shows that no universal formula $N = k + 2\sqrt{k - \mu}$ governs the scalar N in $B^2 = NI$: the strongly-regular-graph mechanism behind $13 = 7 + 2 \cdot 3$ is special to the $G_2(2)$ geometry. Companion GAP scripts: `hecke_irreps.g`, `hecke_irreps2.g`, `hecke_spherical.g`.

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1 Method

1.1 The Hecke algebra of a coset pair

Let G be a finite group, $H \le G$, $n = [G : H]$, and let $\pi = \mathrm{ind}_H^G \mathbf{1}$ be the permutation character of G on $\mathbb{C}[G/H]$. The *Hecke algebra* is

$$\mathcal{H}(G,H) \;=\; \mathrm{End}_G\,\mathbb{C}[G/H] \;\cong\; \bigoplus_{\chi} M_{m_\chi}(\mathbb{C}), \qquad m_\chi = \langle \pi, \chi \rangle.$$

A basis is given by the *orbital matrices* $A_1, \ldots, A_r$: the 0/1 matrices of the orbits of G on $(G/H)^2$, equivalently the adjacency matrices of the suborbits of the point stabilizer. The rank satisfies $r = \sum_\chi m_\chi^2$, and (G,H) is a *Gelfand pair* iff every $m_\chi \le 1$, iff $\mathcal{H}$ is commutative.

By double-centralizer duality the irreducible representations of $\mathcal{H}(G,H)$ are indexed by the constituents χ of π; the Hecke irrep attached to χ has dimension m_χ (the multiplicity, *not* the degree $\chi(1)$), and occurs in $\mathbb{C}[G/H]$ with multiplicity $\chi(1)$. For $m_\chi = 1$ the irrep is the *spherical function* $\omega_\chi : T_i \mapsto \omega_\chi(T_i) \in \mathbb{C}$.

1.2 Exact extraction of the m-dimensional irreps

All matrices in this paper were obtained by the following method, which stays exact over cyclotomic fields and avoids the MeatAxe.

1. Compute the isotypic projector $P_\chi = \frac{\chi(1)}{|G|} \sum_{g \in G} \chi(g^{-1})\, \pi(g)$ via class-sum matrices. Its image W is the χ-isotypic component, $W \cong U_\chi \otimes \mathbb{C}^{\chi(1)}$ where U_χ is the m-dimensional Hecke irrep (A_i acts as $\rho(T_i) \otimes 1$).

2. Find an eigenvector: pick an orbital matrix A_a and an eigenvalue λ of degree ≤ 2 over $\mathbb{Q}$ such that $\dim\big(\ker(A_a - \lambda) \cap W\big) = \chi(1)$. This dimension count forces λ to be a *simple* eigenvalue of $\rho(T_a)$ on U_χ, so every vector of the intersection is a pure tensor $e_\lambda \otimes u$.

3. Spin: the closure of one such vector under $\{A_i\}$ is then a *single* m-dimensional irreducible $\mathcal{H}$-module $U_\chi \otimes u$. Restricting each A_i to it gives exact $m \times m$ matrices.

4. Normalize: change to an eigenbasis of one generator, and for $m = 2$ rescale so that some off-diagonal generator becomes symmetric.

5. Verify: the algebra generated by $\{\rho(T_i)\}$ has dimension m^2, and $\rho\big(\sum_i T_i\big) = 0$ (the all-ones matrix dies on every nontrivial block). Both checks passed in every case below.

1.3 Conventions and notation

T_1 is the identity orbital; suborbits are sorted by length (this numbering differs from `MEINHOLD-HECKE.md` / `why13.tex`; the dictionary is given in §6). Matrices act on row vectors. Recurring quadratic irrationalities:

symbol	value	satisfies	field
α	$\frac{-1+\sqrt{-7}}{2} = \zeta_7 + \zeta_7^2 + \zeta_7^4$	$\alpha^2 + \alpha + 2 = 0,\ \alpha\bar{\alpha} = 2$	$\mathbb{Q}(\sqrt{-7})$
η_0	$\frac{-1+\sqrt{13}}{2} = \sum_{a \in \mathrm{QR}_{13}} \zeta_{13}^a$	$\eta^2 + \eta - 3 = 0,\ \eta_0\eta_1 = -3$	$\mathbb{Q}(\sqrt{13})$
η_1	$\frac{-1-\sqrt{13}}{2}$	$\eta_0 - \eta_1 = \sqrt{13}$	$\mathbb{Q}(\sqrt{13})$
$\sqrt{5}$	$(\zeta_5 + \zeta_5^4) - (\zeta_5^2 + \zeta_5^3)$	Gauss sum mod 5	$\mathbb{Q}(\sqrt{5})$

2 The Gelfand backbone: spherical function tables

Every distance-transitive graph in the catalogue yields a Gelfand pair $(\mathrm{Aut}\,\Gamma, \mathrm{Stab}(v))$ whose Hecke algebra is the Bose–Mesner algebra of the distance scheme: rank = diameter +1, and the spherical values $\omega_\chi(T_i)$ on the distance-i orbital are the entries of the P-matrix (graph eigenvalue data). The tables below were computed by `hecke_spherical.g`; rows are spherical functions, columns the orbitals $T_1, \ldots, T_r$ sorted by suborbit length.

Below, each table lists the spherical functions ω_χ (rows, labelled by the constituent χ and its degree) against the orbitals (columns, headed by suborbit lengths). Cyclotomic outputs have been converted to surds; each conversion was verified in GAP ($E(8) - E(8)^3 = \sqrt{2}$, $-E(12)^7 + E(12)^{11} = \sqrt{3}$, $-E(5) + E(5)^2 + E(5)^3 - E(5)^4 = -\sqrt{5}$, etc.).

2.1 Complete graphs and the cube family

K_4 (rank 2; suborbits $1, 3$): $\chi_{\deg 1} : (1, 3)$; $\chi_{\deg 3} : (1, -1)$.

$K_{3,3}$ (rank 3; suborbits $1, 2, 3$):

	1	2	3
deg 1	1	2	3
deg 1	1	2	−3
deg 4	1	−1	0

Cube (rank 4) :

	1	1	3	3
deg 1	1	1	3	3
deg 1	1	−1	−3	3
deg 3	1	1	−1	−1
deg 3	1	−1	1	−1

2.2 Petersen, Heawood, Möbius–Kantor, Pappus, Desargues

Petersen :

	1	3	6
deg 1	1	3	6
deg 4	1	−2	1
deg 5	1	1	−2

Heawood :

	1	3	4	6
deg 1	1	3	4	6
deg 1	1	−3	−4	6
deg 6	1	$-\sqrt{2}$	$\sqrt{2}$	−1
deg 6	1	$\sqrt{2}$	$-\sqrt{2}$	−1

(The identical table holds for $\mathrm{PSL}(2,5)/S_3$ — Petersen as a coset graph — and the Heawood data realizes the eigenvalues $\pm\sqrt{2}$ of the incidence graph of $\mathrm{PG}(2,2)$.)

Möbius–Kantor :

	1	1	2	3	3	6
deg 1	1	1	2	3	3	6
deg 1	1	1	−2	−3	−3	6
deg 3	1	1	−2	1	1	−2
deg 3	1	1	2	−1	−1	−2
deg 4	1	−1	0	$\sqrt{3}$	$-\sqrt{3}$	0
deg 4	1	−1	0	$-\sqrt{3}$	$\sqrt{3}$	0

Pappus :

	1	2	3	6	6
deg 1	1	2	3	6	6
deg 1	1	2	−3	6	−6
deg 4	1	2	0	−3	0
deg 6	1	−1	$\sqrt{3}$	0	$-\sqrt{3}$
deg 6	1	−1	$-\sqrt{3}$	0	$\sqrt{3}$

Desargues :

	1	1	3	3	6	6
deg 1	1	1	3	3	6	6
deg 1	1	−1	−3	3	6	−6
deg 4	1	1	−2	−2	1	1
deg 4	1	−1	2	−2	1	−1
deg 5	1	1	1	1	−2	−2
deg 5	1	−1	−1	1	−2	2

2.3 Dodecahedron and Nauru

Dodecahedron :

	1	1	3	3	6	6
deg 1	1	1	3	3	6	6
deg 3	1	−1	$-\sqrt{5}$	$\sqrt{5}$	2	−2
deg 3	1	−1	$\sqrt{5}$	$-\sqrt{5}$	2	−2
deg 4	1	1	−2	−2	1	1
deg 4	1	−1	0	0	−3	3
deg 5	1	1	1	1	−2	−2

Nauru :

	1	2	3	3	3	6	6
deg 1	1	2	3	3	3	6	6
deg 1	1	2	−3	−3	3	6	−6
deg 3	1	2	−1	3	−1	−2	−2
deg 3	1	2	1	−3	−1	−2	2
deg 4	1	−1	0	0	3	−3	0
deg 6	1	−1	2	0	−1	1	−2
deg 6	1	−1	−2	0	−1	1	2

Note the dodecahedron's Aut-table (here, Aut $= C_2 \times A_5$, rank 6) versus its rank-8 refinement under $\mathrm{PSL}(2,5)$ acting on cosets of C_3 (§3): the smaller group splits two length-6 suborbits into four of length 3 and turns the pair Gelfand → non-Gelfand.

2.4 Tutte–Coxeter, the 12-cage, and $GH(2,2)$

Tutte–Coxeter :

	1	3	6	8	12
deg 1	1	3	6	8	12
deg 1	1	−3	6	8	−12
deg 9	1	−2	1	−2	2
deg 9	1	2	1	−2	−2
deg 10	1	0	−3	2	0

12-cage orbits $= G_2(2)/P_1$:

	1	6	24	32
deg 1	1	6	24	32
deg 14	1	−3	6	−4
deg 21	1	3	0	−4
deg 27	1	−1	−4	4

The Tutte 12-cage is *not* vertex-transitive (orbits $63+63$); both orbits are the point/line actions of $G_2(2)$ on $GH(2,2)$ and produce *literally identical* spherical tables, even though no automorphism exchanges them at $q=2$. The eigenvalue row $(6, 3^{21}, -1^{27} \dots)$ matches `why13.tex` §3.

2.5 Coxeter-candidate coset pairs in $\mathrm{PGL}(2,7)$

$\mathrm{PGL}(2,7)/A_4(n=28)$:

	1	1	4	4	6	12
deg 1	1	1	4	4	6	12
deg 1	1	1	−4	−4	−6	12
deg 6	1	1	$-\sqrt{2}$	$-\sqrt{2}$	$2\sqrt{2}$	−2
deg 6	1	1	$\sqrt{2}$	$\sqrt{2}$	$-2\sqrt{2}$	−2
deg 7	1	−1	−2	2	0	0
deg 7	1	−1	2	−2	0	0

$\mathrm{PGL}(2,7)/D_{12}(n=28)$:

	1		3	6	6	12
deg 1	1		3	6	6	12
deg 6	1	$-1+\sqrt{2}$		$2+\sqrt{2}$	$-2\sqrt{2}$	−2
deg 6	1	$-1-\sqrt{2}$		$2-\sqrt{2}$	$2\sqrt{2}$	−2
deg 7	1		−1	−2	−2	4
deg 8	1		2	−2	1	−2

2.6 Biggs–Smith: $\mathrm{PSL}(2,17)/S_4$, $n = 102$

Rank 8, suborbits $(1,3,6,8,12,24,24,24)$. Three Galois orbits of spherical functions: the trivial one; a $\mathbb{Q}(\sqrt{17})$ pair (deg 9); a cubic triple (deg 16) over $\mathbb{Q}(\zeta_9)^+$; and two rational rows (deg 17, 18).

	1	3	6	8	12	24	24	24
deg 1	1	3	6	8	12	24	24	24
deg 9	1	$\frac{1+\sqrt{17}}{2}$	$\frac{3+\sqrt{17}}{2}$	$1-\sqrt{17}$	4	$-1+\sqrt{17}$	-8	$1-\sqrt{17}$
deg 9	1	$\frac{1-\sqrt{17}}{2}$	$\frac{3-\sqrt{17}}{2}$	$1+\sqrt{17}$	4	$-1-\sqrt{17}$	-8	$1+\sqrt{17}$
deg 16	1	$-\beta-1$	$\beta^2+2\beta-2$	$\beta^2+\beta-4$	$-3\beta^2-\beta+5$	$2\beta^2+\beta-4$	$-2\beta^2+5$	$\beta^2-2\beta$
deg 17	1	0	-3	2	0	6	0	-6
deg 18	1	2	1	2	-2	-6	4	-2

where $\beta = \zeta_9 + \zeta_9^{-1}$ runs over the three roots of $\beta^3 - 3\beta + 1 = 0$ (one row per root; the orbital order of the three length-24 columns is permuted between conjugate rows — see `hecke_spherical.out` for the exact alignment). Substituting $\beta = -(t+1)$ recovers the Biggs–Smith adjacency cubic $t^3 + 3t^2 - 3$: the spectrum $\{3,\ \frac{1\pm\sqrt{17}}{2}^{(9)},\ (t^3+3t^2-3)^{(16\times 3)},\ 0^{(17)},\ 2^{(18)}\}$ is reassembled block by block.

2.7 Two-transitive geometries (rank 2) and $GQ(3,3)$

pair	ω_{triv}	$\omega_{\text{Steinberg-like}}$
$\mathrm{PSL}(3,3)/P_1$ (13 pts PG(2, 3))	$(1,12)$	$(1,-1)$
$\mathrm{PSL}(3,4)/P_1$ (21 pts PG(2, 4))	$(1,20)$	$(1,-1)$
$\mathrm{PSU}(3,3)/H_{28}$ (Hermitian unital)	$(1,27)$	$(1,-1)$
$\mathrm{Sz}(8)/H_{65}$ (Suzuki ovoid)	$(1,64)$	$(1,-1)$

$\mathrm{PSp}(4,3)/P_1$(40 pts of $GQ(3,3)$) :

	1	12	27
deg 1	1	12	27
deg 15	1	-4	3
deg 24	1	2	-3

3 $\mathrm{PSL}(2,5)/C_3$, $n = 20$: the dodecahedral miniature

$\mathcal{H} \cong \mathbb{C}^4 \oplus M_2(\mathbb{C})$, rank 8, suborbits $(1,1,3,3,3,3,3,3)$. The coset graph of the valency-3 orbital T_3 is the *dodecahedron*.

3.1 Spherical functions

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
χ_1 (deg 1)	1	1	3	3	3	3	3	3
χ_2 (deg 3)	1	-1	$-\sqrt{5}$	$\sqrt{5}$	1	-1	1	-1
χ_3 (deg 3)	1	-1	$\sqrt{5}$	$-\sqrt{5}$	1	-1	1	-1
χ_5 (deg 5)	1	1	1	1	-1	-1	-1	-1

3.2 The 2-dimensional irrep (on χ_4, $\deg 4$)

In the eigenbasis of $\rho(T_2)$:

$$\rho(T_1) = I, \quad \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \rho(T_3) = \rho(T_4) = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix},$$

$$\rho(T_5) = \begin{pmatrix} 1/2 & 1 \\ -15/4 & -3/2 \end{pmatrix}, \quad \rho(T_6) = \begin{pmatrix} 1/2 & -1 \\ -15/4 & 3/2 \end{pmatrix}, \quad \rho(T_7) = \begin{pmatrix} 1/2 & -1 \\ 15/4 & -3/2 \end{pmatrix}, \quad \rho(T_8) = \begin{pmatrix} 1/2 & 1 \\ 15/4 & 3/2 \end{pmatrix}.$$

Characteristic polynomials: $\rho(T_2)$: x^2-1; $\rho(T_3) = \rho(T_4)$: $x(x+2)$; $\rho(T_5), \rho(T_7)$: x^2+x+3 (complex, $\det = 3 =$ valency); $\rho(T_6), \rho(T_8)$: $(x-3)(x+1)$.

Proposition 3.1 (Relations)**.** *Set $A = \rho(T_2)$, $C = \rho(T_5)$. Then*

$$A^2 = I, \qquad C^2 + C + 3I = 0, \qquad AC = \rho(T_8), \quad CA = \rho(T_6), \quad ACA = \rho(T_7),$$

and the closest analogue of $B^2 = 13I$ is the shifted square

$$\big(\rho(T_6) - I\big)^2 = 4I.$$

Remark 3.2. The block does not separate T_3 from T_4 ($\rho(T_3) = \rho(T_4)$); they are separated by χ_2, χ_3 via $\pm\sqrt{5}$. Reassembling the spectrum of the symmetric orbital T_3 from all blocks gives $\{3 \mid \pm\sqrt{5} \mid 0, -2 \mid 1\}$ with multiplicities $\{1 \mid 3, 3 \mid 4, 4 \mid 5\}$ — exactly the dodecahedron spectrum $3,\ \sqrt{5}^3,\ 1^5,\ 0^4,\ (-2)^4,\ (-\sqrt{5})^3$. The famous $\pm\sqrt{5}$ of the dodecahedron lives entirely in the *one-dimensional* Hecke irreps (the degree-3 characters of A_5); the 2×2 block contributes only integer eigenvalues.

4 $\mathrm{PSL}(2,7)/C_3$, $n = 56$

$\mathcal{H} \cong \mathbb{C}^3 \oplus M_2 \oplus M_2 \oplus M_3$, rank 20, suborbits $(1, 1, 3^{18})$.

4.1 The $\mathbb{Q}(\sqrt{-7})$ block (on χ_4, $\deg 6$)

Here $\rho(T_2) = I$: the free involution acts trivially on this block. With $\alpha = \frac{-1+\sqrt{-7}}{2}$, the distinct images are

matrix	which T_i	charpoly
$\mathrm{diag}(\alpha, \bar{\alpha})$	T_3, T_5, T_{12}, T_{17}	$x^2 + x + 2$
$\mathrm{diag}(\bar{\alpha}, \alpha)$	T_4, T_6, T_7, T_{14}	$x^2 + x + 2$
$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$	T_8, T_{18}	$(x-3)(x+1)$
$\begin{pmatrix} 1 & \alpha - 1 \\ \bar{\alpha} - 1 & 1 \end{pmatrix}$	T_9, T_{15}	$(x-3)(x+1)$
$\begin{pmatrix} 0 & \bar{\alpha} \\ \alpha & 0 \end{pmatrix}$	$T_{10}, T_{11}, T_{16}, T_{20}$	$x^2 - 2$
$\begin{pmatrix} -1 & -\bar{\alpha} \\ -\alpha & -1 \end{pmatrix}$	T_{13}, T_{19}	$x^2 + 2x - 1$

Proposition 4.1. *Set* $D = \mathrm{diag}(\alpha, \bar{\alpha})$ *and* $S = \left(\begin{smallmatrix} 0 & \bar{\alpha} \\ \alpha & 0 \end{smallmatrix}\right)$. *Then*

$$D^2 + D + 2I = 0, \qquad \boxed{S^2 = 2I}, \qquad SDS^{-1} = \bar{D}, \qquad \rho(T_{13}) = -I - S,$$

where $2 = \alpha\bar{\alpha} = \mathrm{Nm}(\alpha)$ *is the prime* 2, *split in* $\mathbb{Q}(\sqrt{-7})$. *The six matrices above span* $M_2(\mathbb{C})$, *so* $\{D, S\}$ *generate the block. The eigenvalues* $-1 \pm \sqrt{2}$ *of* $\rho(T_{13})$ *exhibit* $\sqrt{2}$ *as the "junior* $\sqrt{13}$*".*

4.2 The integral 3-dimensional block (on χ_5, $\deg 7$)

In the eigenbasis of $\rho(T_2) = \mathrm{diag}(1, -1, -1)$ *all twenty* images are integer matrices. Representative images:

$$\rho(T_3) = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 2 \\ 4 & 0 & 1 \end{pmatrix}, \quad \rho(T_4) = \begin{pmatrix} 1 & 0 & 1 \\ 4 & 1 & 0 \\ 2 & 2 & 0 \end{pmatrix}, \quad \rho(T_8) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & -3 \end{pmatrix}, \quad \rho(T_{10}) = \begin{pmatrix} -1 & -1 & 1 \\ 2 & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix},$$

$$\rho(T_{11}) = \begin{pmatrix} -1 & 1 & -1 \\ 2 & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix}, \quad \rho(T_{13}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 2 \\ 0 & -2 & 1 \end{pmatrix},$$

the remaining ones being sign/transpose variants (full list in `hecke_irreps.out`). Characteristic polynomials:

charpoly	which T_i
$x^3 - 2x^2 - x - 6$	T_3, T_4, T_{12}, T_{14}
$x^3 + x - 6$ (irreducible over $\mathbb{Q}$: genuinely cubic eigenvalues)	T_5, T_6, T_7, T_{17}
$(x+3)(x-1)(x+1)$	T_8, T_{15}
$(x-3)(x+1)^2$	T_9, T_{18}
$x(x^2 + x + 4)$	T_{10}, T_{16}
$x(x^2 + x - 4)$ eigenvalues $0,\ \frac{-1\pm\sqrt{17}}{2}$	T_{11}, T_{20}
$(x+1)(x^2+3)$ eigenvalues $-1,\ \pm i\sqrt{3}$	T_{13}, T_{19}

4.3 The rational block (on χ_6, $\deg 8$)

In the eigenbasis of $\rho(T_2) = \mathrm{diag}(1,-1)$:

matrix	which T_i	charpoly
$\begin{pmatrix} -1/2 & 1 \\ -7/4 & -1/2 \end{pmatrix}$	T_3, T_{14}	x^2+x+2
$\begin{pmatrix} -1/2 & -1 \\ 7/4 & -1/2 \end{pmatrix}$	T_4, T_{12}	x^2+x+2
$\begin{pmatrix} -1/2 & 1 \\ 7/4 & 1/2 \end{pmatrix}$	T_5, T_7	x^2-2
$\begin{pmatrix} -1/2 & -1 \\ -7/4 & 1/2 \end{pmatrix}$	T_6, T_{17}	x^2-2
$\mathrm{diag}(-1,1)$	T_8, T_{15}	x^2-1
$-I$	T_9, T_{18}	$(x+1)^2$
$\begin{pmatrix} 1/2 & 1 \\ -7/4 & -3/2 \end{pmatrix}$	T_{10} (T_{16} variant)	$x^2+x+1 \quad (\rho^3 = I)$
$\begin{pmatrix} 1/2 & -1 \\ -7/4 & 3/2 \end{pmatrix}$	T_{11} (T_{20} variant)	$x^2-2x-1 \quad (1 \pm \sqrt{2})$
$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$	T_{13}, T_{19}	$x(x-2)$

Proposition 4.2. *With $B = \rho(T_5)$: $\boxed{B^2 = 2I}$ — the same norm-2 phenomenon as in χ_4, now realized over $\mathbb{Q}$. Moreover $\rho(T_{10})$ has* finite order 3 *(charpoly x^2+x+1): a Hecke generator acting as a rotation. The eigenvalues $1 \pm \sqrt{2}$ recur: the $\mathbb{Z}[\sqrt{2}]$ thread runs through both 2-dimensional blocks.*

5 F26A, $n = 26$: Galois-conjugate blocks over $\mathbb{Q}(\sqrt{13})$

$\mathrm{Aut} = C_{13} \rtimes C_6$ (Frobenius, sharply arc-transitive), $\mathcal{H} \cong \mathbb{C}^2 \oplus M_2 \oplus M_2$, rank 10, suborbits $(1,1,3^8)$.

5.1 Spherical functions

χ_1: $(1,1,3,3,3,3,3,3,3,3)$; χ_2: $(1,-1,-3,3,-3,3,-3,3,-3,3)$.

5.2 The 2-dimensional irreps

The two M_2 blocks (on the two degree-6 characters of $C_{13} \rtimes C_6$) are *Galois conjugate over* $\mathbb{Q}(\sqrt{13})$: swapping $\eta_0 \leftrightarrow \eta_1$ passes between them. On χ_7, in the eigenbasis of $\rho(T_2) = \mathrm{diag}(1,-1)$ and with $\gamma := 2\eta_0 + \frac{5}{4}\eta_1$:

$$\rho(T_3) = \begin{pmatrix} \eta_0/2 & 1 \\ -\gamma & -\eta_0/2 \end{pmatrix}, \quad \rho(T_4) = \begin{pmatrix} \eta_1/2 & -(\eta_0+2\eta_1) \\ -\sqrt{13}/4 & \eta_1/2 \end{pmatrix}, \quad \rho(T_5) = \begin{pmatrix} \eta_0/2 & -1 \\ \gamma & -\eta_0/2 \end{pmatrix},$$

$$\rho(T_6) = \begin{pmatrix} \eta_0/2 & -1 \\ -\gamma & \eta_0/2 \end{pmatrix}, \quad \rho(T_7) = \begin{pmatrix} \eta_1/2 & \eta_0+2\eta_1 \\ -\sqrt{13}/4 & -\eta_1/2 \end{pmatrix}, \quad \rho(T_8) = \begin{pmatrix} \eta_1/2 & \eta_0+2\eta_1 \\ \sqrt{13}/4 & \eta_1/2 \end{pmatrix},$$

$$\rho(T_9) = \begin{pmatrix} \eta_1/2 & -(\eta_0+2\eta_1) \\ \sqrt{13}/4 & -\eta_1/2 \end{pmatrix}, \quad \rho(T_{10}) = \begin{pmatrix} \eta_0/2 & 1 \\ \gamma & \eta_0/2 \end{pmatrix}.$$

Proposition 5.1 (F26A relations)**.** *With* $2-\eta_0=\frac{5-\sqrt{13}}{2}$ *and* $2-\eta_1=\frac{5+\sqrt{13}}{2}$*:*

T_i	*relation*
T_3, T_5	$T^2=\frac{5-\sqrt{13}}{2}\,I$
T_7, T_9	$T^2=\frac{5+\sqrt{13}}{2}\,I$
T_4, T_8	$T^2=\eta_1 T-(2-\eta_1)I$
T_6, T_{10}	$T^2=\eta_0 T-(2-\eta_0)I$

The traceless generators square to the scalars $N_\pm=\frac{5\mp\sqrt{13}}{2}$ *with*

$$N_+N_-=3\ (\textit{the valency}),\qquad N_++N_-=5.$$

Remark 5.2. This 13 is the *same numeral* as Meinhold's but enters by a different door: here $26=2\cdot 13$ and $\sqrt{13}$ arises from the quadratic Gauss periods η_0,η_1 of $\mathbb{Q}(\zeta_{13})$, whereas in §6 it arises from the strongly regular graph identity $13=7+2\cdot 3$. The two Galois-conjugate blocks share the rational invariants $(3,5)$.

6 $G_2(2)/\mathrm{PSL}(3,2)$ (Meinhold-72): cross-check

$\mathcal{H}\cong\mathbb{C}^2\oplus M_2\oplus\mathbb{C}^2$, rank 8, suborbits $(1,1,7,7,7,7,21,21)$. Numbering dictionary to `why13.tex/MEINHOLD-HECKE.md`: here T_2 is the free involution "A"; T_4/T_6 is the symmetric valency-7 pair "B/B'" (old T_3/T_5); T_3/T_5 is the directed pair (old T_4/T_6).

6.1 Spherical functions

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
χ_1	1	1	7	7	7	7	21	21
χ_2	1	−1	7	−7	7	−7	−21	21
χ_{21}	1	−1	1	−1	1	−1	3	−3
χ'_{21}	1	1	1	1	1	1	−3	−3

6.2 The 2-dimensional irrep (on χ_{14})

In the eigenbasis of $\rho(T_2)$:

$$\rho(T_2)=\begin{pmatrix}1&0\\0&-1\end{pmatrix},\quad \rho(T_3)=\begin{pmatrix}-2&1\\-9&-2\end{pmatrix},\quad \rho(T_4)=\begin{pmatrix}-2&-1\\-9&2\end{pmatrix},\quad \rho(T_5)=\begin{pmatrix}-2&-1\\9&-2\end{pmatrix},$$

$$\rho(T_6)=\begin{pmatrix}-2&1\\9&2\end{pmatrix},\quad \rho(T_7)=\begin{pmatrix}3&0\\0&-3\end{pmatrix},\quad \rho(T_8)=3I.$$

Charpolys: $\rho(T_3),\rho(T_5)$: $x^2+4x+13$ (the $-2\pm 3i$ pair, $|-2+3i|^2=13$); $\rho(T_4),\rho(T_6)$: x^2-13; $\rho(T_7)$: x^2-9; $\rho(T_8)$ scalar.

Theorem 6.1 (Independent re-derivation of the why-13 algebra)**.** *With* $A=\rho(T_2)$ *and* $B=\rho(T_4)$*:*

$$A^2=I,\qquad \boxed{B^2=13I},\qquad AB+BA=-4I,$$

and the directed pair is $BA=\rho(T_3)$, $AB=\rho(T_5)$*. This reproduces* why13.tex *Theorem 8.1 by isotypic projection, a method sharing nothing with the original orbital path-counting derivation.*

7 $\mathrm{PSL}(2,13)/S_3$, $n = 182$

$\mathcal{H} \cong \mathbb{C}^4 \oplus M_2^{\oplus 3} \oplus M_3^{\oplus 2}$, rank $34 = 4 + 3\cdot 4 + 2\cdot 9$, suborbits $(1, 1, 3^4, 6^{28})$. All matrices below from `hecke_irreps2.out` (both S_3-classes give equivalent data; we use class #1). Set $c_k = \zeta_7^k + \zeta_7^{-k}$, so $c_1 + c_2 + c_3 = -1$ and each c_k satisfies $x^3 + x^2 - 2x - 1 = 0$.

7.1 One-dimensional irreps

Besides the trivial and sign rows, the two degree-7 constituents give spherical functions with values in $\mathbb{Q}(\sqrt{13})$ via the *same* Gauss periods η_0, η_1 as F26A (§5): e.g. $\omega_{\chi_2}(T_3) = -\eta_0 = \frac{1-\sqrt{13}}{2}$ and length-6 orbitals take values such as $2 - \eta_1 = \frac{5+\sqrt{13}}{2}$ (full vectors in `hecke_irreps2.out`). Both $\sqrt{13}$'s — F26A's and this one — come from $\mathbb{Q}(\zeta_{13})$, unlike Meinhold's.

7.2 The three Galois-conjugate M_2 blocks (on χ_4, χ_5, χ_6, deg 12)

The three degree-12 characters of $\mathrm{PSL}(2,13)$ are conjugate under $\mathrm{Gal}(\mathbb{Q}(\zeta_7)^+/\mathbb{Q}) \cong C_3$, and so are the three Hecke blocks: the substitution $c_k \mapsto c_{2k}$ maps the χ_4 block to the χ_6 block to the χ_5 block. (Verified on the charpolys of the valency-3 generator: $x^2 - (c_2{+}c_3)x - (c_1{+}2c_2{+}3c_3)$ for χ_4, $x^2 - (c_1{+}c_3)x - (3c_1{+}c_2{+}2c_3)$ for χ_6, $x^2 - (c_1{+}c_2)x - (2c_1{+}3c_2{+}c_3)$ for χ_5.) It therefore suffices to give the χ_4 block; in the eigenbasis of $\rho(T_2)$,

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \rho(T_3) = \begin{pmatrix} \frac{1}{2}c_1 + \frac{1}{2}c_2 + c_3 & 1 \\ \frac{3}{2}c_1 + \frac{7}{4}c_2 + \frac{11}{4}c_3 & -\frac{1}{2}c_1 + \frac{1}{2}c_2 \end{pmatrix},$$

and $\{\rho(T_2), \rho(T_3)\}$ already generate the block ($\rho(T_3)$ is non-diagonal with distinct eigenvalues). The remaining 31 images are in `hecke_irreps2.out` (lines 60–307). Unlike all previous M_2 blocks, no orbital here squares to a rational scalar: the entries genuinely need the cubic field $\mathbb{Q}(\zeta_7)^+$.

7.3 The integral Steinberg block (on χ_7, deg 13)

All 34 images are *integer* matrices. In the eigenbasis of $\rho(T_2) = \mathrm{diag}(1, 1, -1)$:

$$\rho(T_3) = \begin{pmatrix} 2 & -1 & -2 \\ 2 & -2 & -3 \\ 1 & 0 & 0 \end{pmatrix}, \quad \rho(T_4) = \begin{pmatrix} 2 & -1 & 2 \\ 2 & -2 & 3 \\ 1 & 0 & 0 \end{pmatrix}, \quad \rho(T_7) = \begin{pmatrix} 1 & 0 & 3 \\ 3 & -1 & 8 \\ 0 & 1 & 3 \end{pmatrix}, \quad \rho(T_{21}) = \begin{pmatrix} -2 & 1 & -3 \\ -5 & 4 & -1 \\ 3 & -1 & -2 \end{pmatrix}.$$

Characteristic polynomial highlights:

charpoly	which T_i
$x^3 + 1 \implies \boxed{\rho(T_3)^3 = -I}$ (finite order 6!)	T_3, T_5
$x^3 - 4x - 1$ (irreducible, totally real)	T_4, T_6
$x^3 + x - 26 \quad (26 = 2 \cdot 13 = n/7)$	T_{21}, T_{26}
$x^3 - 3x^2 - 9x + 2,\ x^3 + 3x^2 + 7x - 2,\ \ldots$	(rest: see `.out`)

Note the echo of $\mathrm{PSL}(2,7)/C_3$'s integral 3-dim block, whose analogous cubic was $x^3 + x - 6$; here it is $x^3 + x - 26$.

7.4 The $\deg 14$ block: a cube root of 5

On χ_9 ($\deg 14$), in the eigenbasis of $\rho(T_2) = \mathrm{diag}(1, 1, -1)$:

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{11}{4} \\ 2 & 1 & 3 \\ 1 & 0 & -\frac{3}{2} \end{pmatrix}, \qquad \text{charpoly } x^3 - 5, \qquad \boxed{\rho(T_3)^3 = 5\,I}.$$

$\rho(T_5)$ satisfies the same relation; $\rho(T_4), \rho(T_6)$ have charpoly $x^3 - 3x^2 - x + 5$. This is the first *cubic* analogue of $B^2 = 13I$: a traceless valency-3 orbital whose *cube* is the scalar 5. The eigenvalues are $\sqrt[3]{5}\,\zeta_3^j$ — the Hecke algebra of a cubic graph manufacturing a cube root of 5.

8 Omitted large cases

Two non-Gelfand pairs of the census are omitted here: $\mathrm{PSL}(2, 31)/S_4$ ($n = 620$, rank 33, six M_2 blocks — the first failure of Gelfand-ness in the S_4 tower; its 1-dimensional spherical values live in $\mathbb{Q}(\sqrt{2})$) and $\mathrm{PSL}(2, 19)/S_3$ ($n = 570$, rank 101, three M_4 blocks). The extraction machinery is in place (`hecke_irreps3.g`, which restricts the orbital matrices to the χ-isotypic subspace before the eigenvalue search); these can be run when wanted.

9 The pattern: which N appears in $B^2 = NI$?

algebra	traceless generator	$B^2 = N\,I$	where N lives
$G_2(2)/L_2(7)$, χ_{14}	valency-7 symmetric	$N = 13 = 7 + 2 \cdot 3$	$\mathbb{Z}$; $\lvert -2 \pm 3i\rvert^2 = 13$ in $\mathbb{Z}[i]$
$\mathrm{PSL}(2, 7)/C_3$, χ_4, χ_6	valency-3 (two blocks)	$N = 2$	$\mathbb{Z}$; $2 = \mathrm{Nm}\left(\frac{-1+\sqrt{-7}}{2}\right)$
F26A, χ_7/χ_8	valency-3	$N = \frac{5 \mp \sqrt{13}}{2}$	$\mathbb{Z}[\eta]$; $\mathrm{Nm}(N) = 3 =$ valency
$\mathrm{PSL}(2, 5)/C_3$, χ_4	$(\rho(T_6) - I)$ shifted	$N = 4$	$\mathbb{Z}$; complex pair has $\det = 3$

In every case the directed/complex orbitals have determinant or norm tied to the valency, while the symmetric generator squares to an N measuring the interference between the two chirality classes; for Meinhold-72 the precise statement is $NN^{\mathsf{T}} = 7I + 2A_{\mathrm{SRG}(36,21,12,12)}$, hence $13 = 7 + 2 \cdot 3$.

Remark 9.1 (No universal formula). There is no formula $N = k + 2\sqrt{k - \mu}$ valid across cases: $\mathrm{PSL}(2, 7)/C_3$ has $N = 2 < k = 3$, so the strongly-regular-graph mechanism is special to the Meinhold geometry. The general synthetic meaning of N remains open (why13.tex §9).

Side findings worth keeping: a Hecke generator of finite order 3 ($\mathrm{PSL}(2, 7)/C_3$, χ_6); genuinely cubic eigenvalues ($x^3 + x - 6$) in the 3-dimensional irrep; $\sqrt{17}$ in the same block; the dodecahedron's $\sqrt{5}$ living entirely in the 1-dimensional irreps.